Olympiad Test Paper — Mathematics (Class 7) — Paper 3

Chapter 15: Finding the Unknown

Paper 3 — (progressively harder)

Subject	Mathematics (NCERT Class 7)
Chapter	15 — Finding the Unknown
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: **+4** for each correct answer · **−1** for each wrong answer · **0** if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Most pitfalls come from dropped signs when distributing or removing nested brackets, treating an average of two speeds as the arithmetic mean instead of total-distance over total-time, and off-by-one slips when forming consecutive-integer or reversed-digit equations. Do not write on this paper — mark answers on your answer sheet.

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1. Solve: $5(2x - 3) - 2(3x - 4) = 2x + 5$.

(A) 6 (B) 3
(C) 1 (D) −6

2. A man is 24 years older than his son. In 2 years his age will be twice the age his son will be at that time. The son's present age (in years) is:

(A) 24 (B) 22
(C) 26 (D) 20

3. Solve: $3(2x - 1) - 4 = 2(x + 3) + 2x$.

(A) 6 (B) 6.5
(C) 13 (D) 1

4. A two-digit number is 5 times the sum of its digits. The only such number whose digits differ by 1 is:

(A) 45 (B) 54
(C) 27 (D) 18

5. Solve: $7x - 2(3x - 5) = 4 - (x - 6)$.

(A) 0 (B) 10

(C) −5 (D) 5

6. The perimeter of a triangle is 60 cm. The second side is twice the first, and the third is 8 cm more than the first. The longest side (in cm) is:

(A) 13 (B) 26

(C) 21 (D) 34

7. Solve: $(3x + 1)/4 - (x - 2)/2 = 1$.

(A) 1 (B) 3

(C) 5 (D) −1

8. A purse has ₹2 and ₹5 coins only. There are 30 coins in all worth ₹99. The number of ₹5 coins is:

(A) 17 (B) 15

(C) 20 (D) 13

9. Solve: $2x - [3 - (x - 4)] = 5x - 13$.

(A) −3 (B) 7

(C) 1 (D) 3

10. A cyclist rides 60 km out at 20 km/h and returns the same 60 km at 30 km/h. The average speed for the whole trip (km/h) is:

(A) 25 (B) 24

(C) 50 (D) 26

11. A mother is 30 years older than her daughter. In 5 years the mother's age will be three times the daughter's age then. The daughter's present age (in years) is:

(A) 10 (B) 5

(C) 15 (D) 8

12. A father is currently 3 times as old as his son. Five years ago the father was 4 times as old. The son's present age (in years) is:

(A) 15 (B) 10

(C) 20 (D) 12

13. Solve: $x/3 + (x + 1)/2 = (x - 1)/6$.

(A) −1 (B) 1

(C) −4 (D) 4

14. The sum of three consecutive multiples of 4 is 84. The largest of them is:

(A) 28 (B) 24

(C) 36 (D) 32

15. Solve: $11 - 4x = 4x - 3$.

(A) 1 (B) 2
(C) −1.75 (D) 1.75

16. If $6x - 5 = 4x + 9$, then the value of $3x - 2$ is:

(A) 7 (B) 21
(C) 5 (D) 19

17. Translate: 'Twice the result of subtracting 3 from a number n equals 5 more than the number.' The equation is:

(A) $2n - 3 = n + 5$ (B) $2(n - 3) = n - 5$
(C) $2(n - 3) = n + 5$ (D) $2(n + 3) = n + 5$

18. A rectangle has length $(2x + 3)$ cm and breadth $(x - 1)$ cm. If the length is 10 cm more than the breadth, the perimeter (in cm) is:

(A) 40 (B) 20
(C) 44 (D) 48

19. Solve: $6x - 5(x - 3) = 2(x - 4) - 7$.

(A) −30 (B) 30
(C) 10 (D) 8

20. Solve: $(x + 2)/(x - 1) = 3/2$.

(A) 5 (B) 1
(C) 7 (D) −8

21. A boat goes 36 km downstream in 3 hours and the same 36 km upstream in 6 hours. The speed of the stream (in km/h) is:

(A) 9 (B) 6
(C) 3 (D) 1.5

22. Pipe A fills a tank in 6 hours and pipe B in 12 hours. Working together, the time (in hours) to fill the tank is:

(A) 9 (B) 18
(C) 4 (D) 3

23. Solve: $2(3x - 1) - 3(x - 4) = 4x + 1$.

(A) −9 (B) 3
(C) 1 (D) 9

24. A boy's age now is one-fourth of his father's age. In 12 years his age will be one-half of his father's age then. The father's present age (in years) is:

(A) 12 (B) 36
(C) 24 (D) 48

25. For which value of k does the equation $2(x - k) = 3x - 8$ have the solution $x = 2$?

(A) 1	(B) −1
(C) 2	(D) 3

26. The sum of three consecutive odd integers is 3 more than twice the smallest. The middle integer is:

(A) −3	(B) −1
(C) 1	(D) 3

27. ₹300 is divided among A, B, C so that A gets ₹20 more than B, and C gets twice B's share. B's share (in ₹) is:

(A) 90	(B) 70
(C) 140	(D) 60

28. Each interior angle of a regular polygon is 144°. The number of sides is:

(A) 8	(B) 10
(C) 12	(D) 9

29. Solve: $(2x - 1)/5 + (x + 2)/2 = (3x + 4)/2$.

(A) −2	(B) 2
(C) 3	(D) 1

30. A shopkeeper sold 40 items, some at ₹12 and the rest at ₹20, taking ₹624 in all. The number of ₹20 items sold was:

(A) 22	(B) 18
(C) 20	(D) 16

31. A train of length L metres passes a pole in 9 seconds and a 90-metre platform in 18 seconds at the same speed. Then L equals:

(A) 180	(B) 45
(C) 120	(D) 90

32. Translate: 'A number n, when its triple is reduced by 7, gives the same value as adding 11 to the number.' The equation is:

(A) $3n + 7 = n + 11$	(B) $3n - 7 = n + 11$
(C) $3(n - 7) = n + 11$	(D) $3n - 7 = n - 11$

33. A two-digit number exceeds 4 times the sum of its digits by 3, and the units digit is 1 more than the tens digit. The number is:

(A) 34	(B) 45
(C) 12	(D) 23

34. Solve: $8x - 3(2x - 7) = 4(x + 2) - 9$.

(A) 11	(B) −11
(C) −5	(D) 5

35. The fraction whose numerator is 3 less than its denominator becomes 4/5 when both numerator and denominator are increased by 2. The original fraction is:

(A) 7/10 (B) 13/16

(C) 10/13 (D) 9/12

36. Solve: $1/(2x - 3) = 2/(3x + 1)$.

(A) 7 (B) −7

(C) 5 (D) 1

37. Solve: $2(3x - 1) - (4x - 7) = x + 8$.

(A) −3 (B) 3

(C) 1 (D) 7

38. Three friends contribute to a gift. The first pays one-third of the total, the second pays one-fourth, and the third pays the remaining ₹250. The total cost (in ₹) is:

(A) 500 (B) 600

(C) 750 (D) 450

39. If $7x - 4 = 5x + 8$, then the value of $x^2 - 3$ is:

(A) 6 (B) 9

(C) 13 (D) 33

40. Solve: $(3x + 2)/5 = (2x - 1)/3$.

(A) −11 (B) −1

(C) 11 (D) 1

41. A sum of money is shared so that for every ₹2 Asha gets, Bina gets ₹3 and Chetan gets ₹5. If Chetan gets ₹60 more than Asha, the total amount (in ₹) is:

(A) 120 (B) 240

(C) 200 (D) 100

42. If $3x - 2y = 12$ and $x + y = 6$, then the value of x is:

(A) 4.8 (B) 3.6

(C) 4 (D) 6

43. Two cars start 180 km apart and drive toward each other, one at 50 km/h and the other at 40 km/h. The time (in hours) until they meet is:

(A) 2 (B) 2.5

(C) 4.5 (D) 18

44. In a regular polygon, each exterior angle is 24°. The number of sides is:

(A) 12 (B) 15

(C) 10 (D) 18

45. When the digits of a two-digit number are reversed, the new number is 27 more than the original. If the units digit is twice the tens digit, the original number is:

(A) 36 (B) 63
(C) 48 (D) 24

46. A man rows downstream at 10 km/h and upstream at 6 km/h. His rowing speed in still water (km/h) is:

(A) 4 (B) 2
(C) 8 (D) 16

47. Solve: $8 - \{2x - (3 - x)\} = 2$.

(A) −3 (B) 1
(C) 9 (D) 3

48. Solve: $x/2 + (2x - 1)/3 = (5x - 4)/6 + 1$.

(A) −2 (B) 2
(C) 1 (D) 4

49. Solve: $3x - [2 - \{4 - (x - 1)\}] = x + 11$.

(A) −8 (B) 6
(C) 10 (D) 8

50. If $5x - 3 = 2x + 9$, then the value of $(x + 1)/2$ is:

(A) 4 (B) 2
(C) 5 (D) 2.5

51. ₹500 is split among three workers. The second receives 1.5 times the first, and the third receives ₹40 less than the second. The first worker's share (in ₹) is:

(A) 135 (B) 202.5
(C) 162.5 (D) 180

52. Solve: $0.5(x - 2) + 0.25(2x + 4) = 3$.

(A) 6 (B) 1
(C) 3 (D) −3

53. A father's age is 3 more than 4 times his son's age. In 5 years the father will be 3 times as old as his son. The son's present age (in years) is:

(A) 5 (B) 9
(C) 11 (D) 7

54. The sum of four consecutive even integers is 4 more than 5 times the smallest. The largest of them is:

(A) 14 (B) 8
(C) 12 (D) 16

55. In a 40-question test, +3 is given for each correct answer and −1 for each wrong answer, with no skipped questions. A student scores 64. The number of correct answers is:

(A) 24
(B) 20
(C) 26
(D) 28

56. A man rows a boat 12 km downstream and back. Downstream he goes at 6 km/h and upstream at 4 km/h. His total time for the round trip (in hours) is:

(A) 4.8
(B) 2.4
(C) 5
(D) 6

57. ₹720 is shared among A, B and C so that A gets ₹30 more than B, and C gets ₹30 more than A. C's share (in ₹) is:

(A) 240
(B) 210
(C) 270
(D) 300

58. The denominator of a fraction exceeds its numerator by 4. If 3 is added to both, the fraction becomes 2/3. The original numerator is:

(A) 9
(B) 5
(C) 7
(D) 1

59. How many values of x satisfy $4(x - 1) + 3 = 2(2x - 5) + 9$?

(A) exactly one
(B) none
(C) infinitely many (every number)
(D) exactly two

60. Solve: $2[3 - (x - 4)] = 3(x + 2) - x$.

(A) −4
(B) 5
(C) 2
(D) 1

Solutions — Mathematics (Class 7), Chapter 15: Finding the Unknown — Paper 3

Paper 3 — (progressively harder)

Marking: **+4** correct, **−1** wrong, **0** unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	11	A	21	C	31	D	41	C	51	A
2	B	12	A	22	C	32	B	42	A	52	C
3	B	13	A	23	D	33	D	43	A	53	D
4	A	14	D	24	C	34	A	44	B	54	A
5	A	15	D	25	D	35	C	45	A	55	C
6	B	16	D	26	B	36	A	46	C	56	C
7	D	17	C	27	B	37	B	47	D	57	C
8	D	18	A	28	B	38	B	48	B	58	B
9	D	19	B	29	A	39	D	49	D	59	C
10	B	20	C	30	B	40	C	50	D	60	C

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (A) Expand both brackets: $10x - 15 - 6x + 8 = 2x + 5$. Combine the left side: $4x - 7 = 2x + 5$. Transpose: $4x - 2x = 5 + 7$, so $2x = 12$ and $x = 6$. Check: left = $5(9) - 2(14) = 45 - 28 = 17$ and right = $12 + 5 = 17$. Halving the answer gives 3; mishandling $-2(3x - 4)$ as $-6x - 8$ corrupts the constant toward 1; flipping the final sign gives −6.

2. (B) Let the son be s now, so the man is $s + 24$. In 2 years the son is $s + 2$ and the man is $s + 26$, and the condition is $s + 26 = 2(s + 2)$. Expand: $s + 26 = 2s + 4$, so $26 - 4 = 2s - s$ and $s = 22$. Check: son 22, man 46; in 2 years they are 24 and 48, and $48 = 2 \times 24$. Comparing present ages $s + 24 = 2s$ gives the trap 24; forgetting to add 2 to the son inside the bracket gives $s + 26 = 2s \rightarrow 26$; doubling the gap gives 20.

3. (B) Expand: $6x - 3 - 4 = 2x + 6 + 2x$, i.e. $6x - 7 = 4x + 6$. Transpose: $6x - 4x = 6 + 7$, so $2x = 13$ and $x = 6.5$. Check: left = $3(12) - 4 = 36 - 4 = 32$ and right = $2(9.5) + 13 = 19 + 13 = 32$. Combining $6x - 3 - 4$ as $6x + 7$ gives a wrong 6; dropping a term on the right gives 13; mis-transposing gives 1.

4. (A) Let the tens digit be t and units u, number 10t + u. The condition 10t + u = 5(t + u) gives 10t + u = 5t + 5u → 5t = 4u. The only single-digit pair satisfying 5t = 4u is t = 4, u = 5, so the number is 45, and its digits differ by 1. Check: 4 + 5 = 9 and 5 × 9 = 45. Its reverse 54 fails 5t = 4u; 27 and 18 also fail the digit-sum-times-5 test.

5. (A) Expand carefully: 7x − 6x + 10 = 4 − x + 6, so x + 10 = 10 − x. Transpose: x + x = 10 − 10, 2x = 0, x = 0. Check: left = 0 − 2(−5) = 10 and right = 4 − (−6) = 10. Dropping the minus on (x − 6) gives x + 10 = 10 + x misread as 10; mishandling −2(3x − 5) as −6x − 10 gives x = −5; a transposition error yields 5.

6. (B) Let the first side be a. The sides are a, 2a, and a + 8, summing to 60: a + 2a + a + 8 = 60 → 4a + 8 = 60 → 4a = 52 → a = 13. The sides are 13, 26, 21, so the longest is 26. Reporting a itself gives 13; reporting the third side a + 8 = 21 mistakes which is longest; adding 8 to the doubled side gives 34.

7. (D) Multiply every term by the LCD 4: (3x + 1) − 2(x − 2) = 4. Expand: 3x + 1 − 2x + 4 = 4 → x + 5 = 4 → x = −1. Check: (−3 + 1)/4 − (−1 − 2)/2 = (−2)/4 − (−3)/2 = −1/2 + 3/2 = 1. Distributing −2(x − 2) as −2x − 4 gives x − 3 = 4 → 7 (toward 5); a sign slip gives 1 or 3.

8. (D) Let the number of ₹5 coins be f, so ₹2 coins number 30 − f. Value: 5f + 2(30 − f) = 99 → 5f + 60 − 2f = 99 → 3f = 39 → f = 13. Check: 13 fives = ₹65 and 17 twos = ₹34, total ₹99. The complement 30 − 13 = 17 is the ₹2 count, a swap trap; splitting equally gives 15; using only the ₹5 value gives 20.

9. (D) Resolve the inner bracket: −(x − 4) = −x + 4, so the square-bracket contents are 3 − x + 4 = 7 − x. Then 2x − (7 − x) = 2x − 7 + x = 3x − 7. The equation is 3x − 7 = 5x − 13. Transpose: −7 + 13 = 5x − 3x → 6 = 2x → x = 3. Check: left = 6 − [3 − (−1)] = 6 − 4 = 2 and right = 15 − 13 = 2. Not flipping the inner sign gives −3; dropping the outer minus gives 7; mis-transposing gives 1.

10. (B) Average speed is total distance over total time, never the mean of the two speeds. Out time = 60/20 = 3 h; return time = 60/30 = 2 h; total 120 km in 5 h, so average = 120/5 = 24 km/h. The naive mean (20 + 30)/2 = 25 is the classic trap; adding the speeds gives 50; a miscount of time gives 26.

11. (A) Let the daughter be d now, so the mother is d + 30. In 5 years the daughter is d + 5 and the mother is d + 35, with d + 35 = 3(d + 5). Expand: d + 35 = 3d + 15 → 35 − 15 = 3d − d → 20 = 2d → d = 10. Check: daughter 10, mother 40; in 5 years 15 and 45, and 45 = 3 × 15. Using present ages d + 30 = 3d gives 15; halving the gap gives 5; mis-adding gives 8.

12. (A) Let the son be s now, so the father is 3s. Five years ago: son s − 5, father 3s − 5, with 3s − 5 = 4(s − 5). Expand: 3s − 5 = 4s − 20 → 20 − 5 = 4s − 3s → s = 15. Check: son 15, father 45; five years ago 10 and 40, and 40 = 4 × 10. Forgetting to subtract 5 from the son gives 3s − 5 = 4s → s = −5; mis-scaling gives 10, 20, or 12.

13. (A) Multiply every term by the LCD 6: $x/3 \times 6 = 2x$, $(x + 1)/2 \times 6 = 3(x + 1)$, $(x - 1)/6 \times 6 = x - 1$. So $2x + 3(x + 1) = x - 1 \rightarrow 2x + 3x + 3 = x - 1 \rightarrow 5x + 3 = x - 1$. Transpose: $5x - x = -1 - 3 \rightarrow 4x = -4 \rightarrow x = -1$. Verify $x = -1$: left $= -1/3 + 0/2 = -1/3$ and right $= (-1 - 1)/6 = -2/6 = -1/3$. A sign flip on the last step gives 1; mis-clearing the denominators gives ±4.

14. (D) Let the multiples be $4n$, $4n + 4$, $4n + 8$. Their sum is $12n + 12 = 84 \rightarrow 12n = 72 \rightarrow n = 6$, giving 24, 28, 32; the largest is 32. Reporting the middle gives 28; the smallest gives 24; adding 4 once more gives 36.

15. (D) Bring the x terms together: $11 + 3 = 4x + 4x \rightarrow 14 = 8x \rightarrow x = 14/8 = 1.75$. Check: left $= 11 - 4(1.75) = 11 - 7 = 4$ and right $= 4(1.75) - 3 = 7 - 3 = 4$. Rounding down gives 1; rounding up gives 2; a sign error gives −1.75.

16. (D) Solve first: $6x - 5 = 4x + 9 \rightarrow 6x - 4x = 9 + 5 \rightarrow 2x = 14 \rightarrow x = 7$. Then $3x - 2 = 21 - 2 = 19$. Stopping at $x = 7$ reports 7; computing $3x$ without the −2 gives 21; a sign slip gives 5.

17. (C) 'Subtracting 3 from n' is $(n - 3)$; 'twice the result' multiplies the whole bracket: $2(n - 3)$. 'Five more than the number' is $n + 5$. So $2(n - 3) = n + 5$. Multiplying only n by 2 drops the bracket; '5 more' is +5 not −5; the inner sign is minus, not plus.

18. (A) The length exceeds the breadth by 10: $(2x + 3) - (x - 1) = 10 \rightarrow x + 4 = 10 \rightarrow x = 6$. Then length = 15 cm, breadth = 5 cm, perimeter $= 2(15 + 5) = 40$ cm. Verify: $15 - 5 = 10$. Using one pair of sides gives 20; mis-solving x gives 44 or 48.

19. (B) Expand: $6x - 5x + 15 = 2x - 8 - 7$, so $x + 15 = 2x - 15$. Transpose: $15 + 15 = 2x - x \rightarrow x = 30$. Check: left $= 180 - 5(27) = 180 - 135 = 45$ and right $= 2(26) - 7 = 52 - 7 = 45$. A sign flip on the bracket gives −30; combining constants wrongly gives 10 or 8.

20. (C) Cross-multiply: $2(x + 2) = 3(x - 1) \rightarrow 2x + 4 = 3x - 3 \rightarrow 4 + 3 = 3x - 2x \rightarrow x = 7$. Check: $(7 + 2)/(7 - 1) = 9/6 = 3/2$. Cross-multiplying the wrong way ($3(x + 2) = 2(x - 1)$) gives $x = -8$; transposition slips give 5 or 1.

21. (C) Downstream speed = 36/3 = 12 km/h = boat + stream; upstream speed = 36/6 = 6 km/h = boat − stream. Subtracting: 2 × stream = 12 − 6 = 6 → stream = 3 km/h (and boat = 9). The boat speed 9 is a trap for the stream; using only one leg gives 6; halving the wrong quantity gives 1.5.

22. (C) Rates add, not times. In one hour A fills 1/6 and B fills 1/12, together $1/6 + 1/12 = 2/12 + 1/12 = 3/12 = 1/4$ of the tank, so the whole tank takes 4 hours. Averaging the times $(6 + 12)/2 = 9$ is the trap; adding times gives 18; a rate slip gives 3.

23. (D) Expand: $6x - 2 - 3x + 12 = 4x + 1$, so $3x + 10 = 4x + 1$. Transpose: $10 - 1 = 4x - 3x \rightarrow x = 9$. Check: left $= 2(26) - 3(5) = 52 - 15 = 37$ and right $= 36 + 1 = 37$. Mishandling $-3(x - 4)$ as $-3x - 12$ gives a sign trap toward −9; other slips give 3 or 1.

24. (C) Let the father's age now be x, so the boy is x/4. In 12 years: boy x/4 + 12, father x + 12, with x/4 + 12 = (x + 12)/2. Multiply by 4: x + 48 = 2(x + 12) → x + 48 = 2x + 24 → 48 − 24 = 2x − x → x = 24. Check: father 24, boy 6; in 12 years 18 and 36, and 18 = 36/2. The boy's future 18 is one trap; 36 is the father's future age; halving and doubling give 12 and 48.

25. (D) Substitute x = 2: 2(2 − k) = 3(2) − 8 → 4 − 2k = −2 → −2k = −6 → k = 3. Verify: 2(2 − 3) = −2 and 3(2) − 8 = −2, equal. A halving slip gives 1; sign errors give −1 or 2.

26. (B) Let the integers be n, n + 2, n + 4. Their sum is 3n + 6, and twice the smallest plus 3 is 2n + 3. Set 3n + 6 = 2n + 3 → n = −3, giving −3, −1, 1; the middle is −1. Check: sum = −3 + −1 + 1 = −3 and 2(−3) + 3 = −3. Reporting the smallest gives −3; the largest gives 1; a sign slip gives 3.

27. (B) Let B = b. Then A = b + 20 and C = 2b, summing to 300: (b + 20) + b + 2b = 300 → 4b + 20 = 300 → 4b = 280 → b = 70. Check: A = 90, B = 70, C = 140, total 300. Reporting A gives 90, C gives 140; ignoring the +20 gives 75 toward 60.

28. (B) Each exterior angle = 180° − 144° = 36°, and exterior angles sum to 360°, so sides = 360/36 = 10. Misreading 180 − 144 as 30 gives 12; 360/45 gives 8; 360/40 gives 9.

29. (A) Multiply every term by the LCD 10: 2(2x − 1) + 5(x + 2) = 5(3x + 4). Expand: 4x − 2 + 5x + 10 = 15x + 20 → 9x + 8 = 15x + 20 → 8 − 20 = 15x − 9x → −12 = 6x → x = −2. Verify x = −2: left = (−5)/5 + (0)/2 = −1 + 0 = −1 and right = (−2)/2 = −1. A sign error gives 2; mis-clearing denominators gives 3 or 1.

30. (B) Let the number of ₹20 items be h, so ₹12 items number 40 − h. Total: 20h + 12(40 − h) = 624 → 20h + 480 − 12h = 624 → 8h = 144 → h = 18. Check: 18 × 20 = 360 and 22 × 12 = 264, total 624. The complement 40 − 18 = 22 is the ₹12 count, a swap trap; splitting evenly gives 20; a sign slip gives 16.

31. (D) Past a pole the train covers L in 9 s, so speed = L/9. Past the platform it covers L + 90 in 18 s, so speed = (L + 90)/18. Equate: L/9 = (L + 90)/18 → 18L = 9(L + 90) → 18L = 9L + 810 → 9L = 810 → L = 90. Check: speed 10 m/s; pole 90/10 = 9 s, platform 180/10 = 18 s. Doubling gives 180; halving gives 45; misadding gives 120.

32. (B) 'Its triple reduced by 7' is 3n − 7. 'Adding 11 to the number' is n + 11. So 3n − 7 = n + 11. 'Reduced by 7' is minus, not plus; the 7 is subtracted after tripling, not inside a bracket; '11 added' is +11, not −11.

33. (D) Let tens t, units u = t + 1, number 10t + u. Condition: 10t + u = 4(t + u) + 3 → 10t + u = 4t + 4u + 3 → 6t − 3u = 3 → 2t − u = 1. Substitute u = t + 1: 2t − (t + 1) = 1 → t − 1 = 1 → t = 2, u = 3, number 23. Check: 4(2 + 3) + 3 = 23. Choosing t = 3 gives 34, t = 4 gives 45 (both fail the +3); reversing the digit relation gives 12.

34. (A) Expand: $8x - 6x + 21 = 4x + 8 - 9$, so $2x + 21 = 4x - 1$. Transpose: $21 + 1 = 4x - 2x \rightarrow 22 = 2x \rightarrow x = 11$. Check: left $= 88 - 3(15) = 88 - 45 = 43$ and right $= 4(13) - 9 = 52 - 9 = 43$. A sign-flip trap gives -11; constant errors give ± 5.

35. (C) Let the denominator be d, numerator $d - 3$, fraction $(d - 3)/d$. After adding 2 to each: $(d - 1)/(d + 2) = 4/5$. Cross-multiply: $5(d - 1) = 4(d + 2) \rightarrow 5d - 5 = 4d + 8 \rightarrow d = 13$, numerator 10, fraction 10/13. Check: $12/15 = 4/5$. Misplacing the gap gives 7/10; off-by-one in d gives 13/16 or 9/12.

36. (A) Cross-multiply: $(3x + 1) = 2(2x - 3) \rightarrow 3x + 1 = 4x - 6 \rightarrow 1 + 6 = 4x - 3x \rightarrow x = 7$. Check: $1/(14 - 3) = 1/11$ and $2/(21 + 1) = 2/22 = 1/11$. A sign error gives -7; transposition slips give 5 or 1.

37. (B) Expand: $6x - 2 - 4x + 7 = x + 8$, so $2x + 5 = x + 8$. Transpose: $2x - x = 8 - 5 \rightarrow x = 3$. Check: left $= 2(8) - 5 = 16 - 5 = 11$ and right $= 3 + 8 = 11$. Not flipping the sign on $(4x - 7)$ gives a wrong 1 or 7; a sign flip gives -3.

38. (B) Let the total be T. The third's share is $T - T/3 - T/4 = T(12 - 4 - 3)/12 = 5T/12 = 250$, so $T = 250 \times 12/5 = 600$. Check: $200 + 150 + 250 = 600$. Misadding the fractions to 1/2 gives 500; arithmetic slips give 750 or 450.

39. (D) Solve: $7x - 4 = 5x + 8 \rightarrow 2x = 12 \rightarrow x = 6$. Then $x^2 - 3 = 36 - 3 = 33$. Stopping at $x = 6$ reports 6; computing only part gives 9; using $2x$ instead of x^2 gives 13.

40. (C) Cross-multiply: $3(3x + 2) = 5(2x - 1) \rightarrow 9x + 6 = 10x - 5 \rightarrow 6 + 5 = 10x - 9x \rightarrow x = 11$. Check: $(33 + 2)/5 = 7$ and $(22 - 1)/3 = 7$. A sign-flip gives -11; transposition slips give ± 1.

41. (C) Let the common multiple be k, so shares are 2k, 3k, 5k. Chetan exceeds Asha by $5k - 2k = 3k = 60 \rightarrow k = 20$. Total $= (2 + 3 + 5)k = 10k = 200$. Check: 40, 60, 100 and $100 - 40 = 60$. Using only Chetan's share gives 100; mis-scaling gives 120 or 240.

42. (A) From $x + y = 6$, $y = 6 - x$. Substitute: $3x - 2(6 - x) = 12 \rightarrow 3x - 12 + 2x = 12 \rightarrow 5x = 24 \rightarrow x = 4.8$. Check: $y = 1.2$ and $3(4.8) - 2(1.2) = 14.4 - 2.4 = 12$. Forgetting to distribute -2 gives $3x - 12 + x = 12 \rightarrow x = 6$; a sign slip gives 3.6; rounding gives 4.

43. (A) Toward each other the closing speed is the sum $50 + 40 = 90$ km/h, so the time to cover 180 km is $180/90 = 2$ hours. Using only one speed gives $180/40 = 4.5$; using the difference 10 km/h gives 18; a half-step error gives 2.5.

44. (B) Exterior angles sum to 360°, so sides $= 360/24 = 15$. Treating 24 as an interior-related angle misapplies the rule, giving 12 or 10; a division slip gives 18.

45. (A) Let tens t, units $u = 2t$, number $10t + u$. Reversed $10u + t = (10t + u) + 27 \rightarrow 9u - 9t = 27 \rightarrow u - t = 3$. With $u = 2t$: $2t - t = 3 \rightarrow t = 3$, $u = 6$, number 36. Check: reversed 63 =

$36 + 27$ and $6 = 2 \times 3$. The reversed number 63 is a trap; $t = 4$, $u = 8$ gives 48 (fails the $+27$); halving gives 24.

46. (C) Downstream speed = still-water + stream = 10; upstream = still-water − stream = 6. Adding: 2 × still-water = 16 → still-water = 8 (and stream = 2). The stream speed 2 is a trap; halving the difference gives 4; adding without halving gives 16.

47. (D) Resolve the inner bracket: $-(3 - x) = -3 + x$, so the brace contents are $2x - 3 + x = 3x - 3$. Then $8 - (3x - 3) = 8 - 3x + 3 = 11 - 3x$. The equation $11 - 3x = 2$ gives $3x = 9$, $x = 3$. Check: $8 - \{6 - 0\} = 8 - 6 = 2$. Not flipping the inner sign gives $8 - (x - 3)$ toward a wrong -3; sign slips give 1 or 9.

48. (B) Multiply every term by the LCD 6: $x/2 \times 6 = 3x$, $(2x - 1)/3 \times 6 = 2(2x - 1)$, $(5x - 4)/6 \times 6 = 5x - 4$, and $1 \times 6 = 6$. So $3x + 2(2x - 1) = 5x - 4 + 6 \rightarrow 3x + 4x - 2 = 5x + 2 \rightarrow 7x - 2 = 5x + 2$. Transpose: $7x - 5x = 2 + 2 \rightarrow 2x = 4 \rightarrow x = 2$. Verify $x = 2$: left $= 1 + 3/3 = 1 + 1 = 2$ and right $= (10 - 4)/6 + 1 = 1 + 1 = 2$. A sign error gives -2; mis-clearing the denominators gives 1 or 4.

49. (D) Work outward. Innermost: $x - 1$. Next: $4 - (x - 1) = 4 - x + 1 = 5 - x$. Next: $2 - \{5 - x\} = 2 - 5 + x = x - 3$. So the left side is $3x - (x - 3) = 3x - x + 3 = 2x + 3$. Equation: $2x + 3 = x + 11 \rightarrow x = 8$. Check: left $= 24 - [2 - \{4 - 7\}] = 24 - [2 - (-3)] = 24 - 5 = 19$ and right $= 8 + 11 = 19$. Sign mishandling of the nested brackets gives -8, 6, or 10.

50. (D) Solve: $5x - 3 = 2x + 9 \rightarrow 3x = 12 \rightarrow x = 4$. Then $(x + 1)/2 = 5/2 = 2.5$. Reporting x itself gives 4; using $x/2$ gives 2; forgetting to divide gives 5.

51. (A) Let the first be f. Second $= 1.5f$, third $= 1.5f - 40$. Sum: $f + 1.5f + (1.5f - 40) = 500 \rightarrow 4f - 40 = 500 \rightarrow 4f = 540 \rightarrow f = 135$. Check: $135 + 202.5 + 162.5 = 500$. Reporting the second gives 202.5; reporting the third gives 162.5; ignoring the -40 gives $4f = 500 \rightarrow 125$ toward 180.

52. (C) Expand: $0.5x - 1 + 0.5x + 1 = 3$, so $x = 3$. Check: $0.5(1) + 0.25(10) = 0.5 + 2.5 = 3$. Mishandling $0.25(2x + 4)$ as $0.25x + 1$ gives a wrong larger root near 6; sign slips give 1 or -3.

53. (D) Let the son be s, so the father is $4s + 3$. In 5 years: son $s + 5$, father $4s + 8$, with $4s + 8 = 3(s + 5)$. Expand: $4s + 8 = 3s + 15 \rightarrow 4s - 3s = 15 - 8 \rightarrow s = 7$. Check: son 7, father 31; in 5 years 12 and 36, and $36 = 3 \times 12$. Forgetting the +3 or mis-adding the +5 gives 5, 9, or 11.

54. (A) Let the integers be n, $n + 2$, $n + 4$, $n + 6$ (even). Their sum is $4n + 12$, and 5 times the smallest plus 4 is $5n + 4$. Set $4n + 12 = 5n + 4 \rightarrow 12 - 4 = 5n - 4n \rightarrow n = 8$, giving 8, 10, 12, 14; the largest is 14. Check: sum 44 and $5(8) + 4 = 44$. Reporting the smallest gives 8; the third gives 12; adding 2 once more gives 16.

55. (C) Let correct = c, wrong = 40 − c. Score: $3c - (40 - c) = 64 \rightarrow 3c - 40 + c = 64 \rightarrow 4c = 104 \rightarrow c = 26$. Check: 26 correct gives 78, 14 wrong gives −14, net 64. Forgetting the penalty gives $3c = 64 \rightarrow$ toward 20; mis-signing gives 24 or 28.

56. (C) Time is distance over speed for each leg. Downstream: $12/6 = 2$ h; upstream: $12/4 = 3$ h; total $= 2 + 3 = 5$ h. Using the average speed $(6 + 4)/2 = 5$ km/h on 24 km gives $24/5 = 4.8$ h, the classic averaging trap; using only one leg's speed on 24 km gives wrong values like 6 or, with $24/10$, 2.4.

57. (C) Let B = b. Then A = b + 30 and C = b + 60, summing to 720: $b + (b + 30) + (b + 60) = 720 \rightarrow 3b + 90 = 720 \rightarrow 3b = 630 \rightarrow b = 210$. So A = 240 and C = 270. Check: 210 + 240 + 270 = 720. Reporting A gives 240, reporting B gives 210; adding one extra ₹30 step gives 300.

58. (B) Let the numerator be n, denominator n + 4. Adding 3 to each: $(n + 3)/(n + 7) = 2/3$. Cross-multiply: $3(n + 3) = 2(n + 7) \rightarrow 3n + 9 = 2n + 14 \rightarrow n = 5$. Check: 5/9 becomes 8/12 = 2/3. Reporting the denominator gives 9; off-by-one gives 7 or 1.

59. (C) Expand both sides: left $= 4x - 4 + 3 = 4x - 1$; right $= 4x - 10 + 9 = 4x - 1$. Both sides reduce to the identical expression $4x - 1$, so $4x - 1 = 4x - 1$ holds for every value of x — infinitely many solutions. Finding one solution and stopping suggests 'exactly one'; reading the constants as unequal suggests 'none'; a linear equation can never give 'exactly two'.

60. (C) Inner bracket on the left: $-(x - 4) = -x + 4$, so $3 - (x - 4) = 7 - x$, and $2(7 - x) = 14 - 2x$. Right side: $3(x + 2) - x = 3x + 6 - x = 2x + 6$. Equation: $14 - 2x = 2x + 6 \rightarrow 14 - 6 = 2x + 2x \rightarrow 8 = 4x \rightarrow x = 2$. Verify x = 2: left $= 2[3 - (-2)] = 2(5) = 10$ and right $= 3(4) - 2 = 12 - 2 = 10$. Not flipping the inner sign gives −4; mis-combining the right side gives 5; a transposition slip gives 1.